

Lab 4 — Written Analysis

DDM501 — AI in DevOps, DataOps, MLOps · Monitoring and Production Deployment
Three traffic profiles analysed against the Model Behaviour dashboard

1. Experimental Setup

The credit risk API was deployed via `docker compose up -d --build` with four containers: the FastAPI service, Prometheus (scraping every 10s), Grafana, and node-exporter. For each traffic profile, the API container was restarted to reset the monitoring window, then 400 requests were sent via `scripts/load_test.py`. The `/monitoring` endpoint was read after each run and the Model Behaviour dashboard was screenshotted from Grafana.

Summary of results

Profile	Score mean	APPROVE	REVIEW	DECLINE	Drift score (max PSI)	Drift status	Fairness gap
Normal	0.2355	77.8%	14.2%	8.0%	0.030	stable	0.019
Drifted (full)	0.5897	15.5%	29.8%	54.8%	4.251	significant	0.038
Unfair	0.3966	52.2%	15.8%	32.0%	0.353	significant	0.722

Per-feature PSI breakdown

Feature	Normal	Drifted	Unfair
LIMIT_BAL	0.015	2.043	0.015
AGE	0.030	1.708	0.030
PAY_0	0.006	0.826	0.279
utilisation_ratio	0.029	3.092	0.029
payment_ratio	0.013	4.251	0.288
max_delay	0.004	0.782	0.353

2. Dashboard Screenshots

2.1 Normal traffic

Is the model still looking at the same world?

Drift score (max PSI)

0.0301

Drift window size

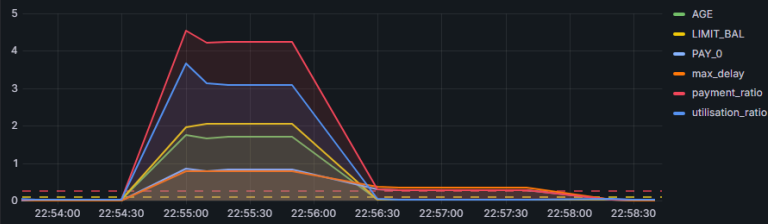
400

Fairness gap

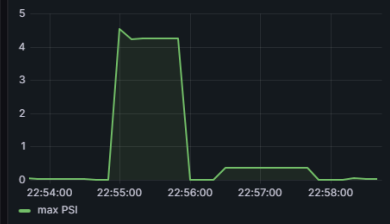
0.0188

Input drift

PSI per feature

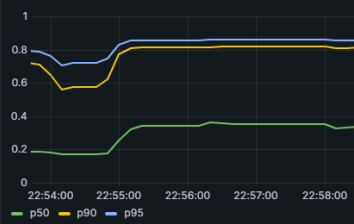


Aggregate drift score

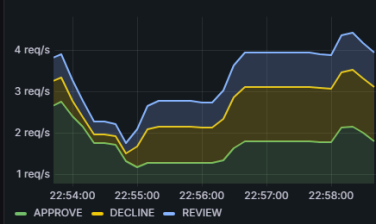


What the model is saying

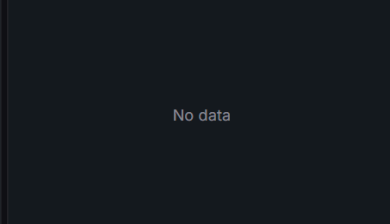
Score distribution (quantiles)



Decision mix

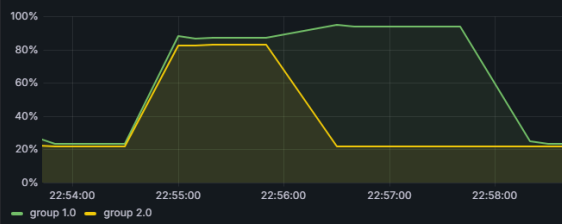


Prediction errors by type

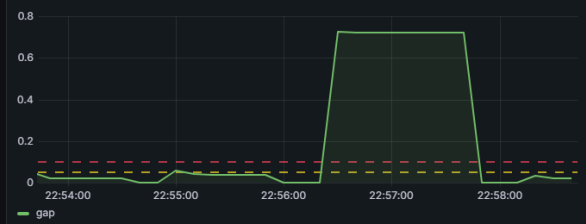


Fairness

Selection rate by group



Fairness gap over time



Cost of explaining

Explanation latency



Explanations rate



Figure 1 — Model Behaviour dashboard after 400 normal requests. All stat panels are green. Drift score 0.030 (stable), fairness gap 0.019.

Under normal traffic, all panels are green. The drift score sits at 0.030 — well below the 0.10 moderate threshold. All six monitored features report PSI values under 0.03, confirming the live window matches the training reference. The fairness gap is 0.019: group 1.0 has a selection rate of 23.4% versus group 2.0 at 21.6%, a difference within statistical noise for 400 requests. The decision mix shows APPROVE dominating at ~78%, with DECLINE at 8% — the expected baseline for this model.

2.2 Drifted traffic

Is the model still looking at the same world?

Drift score (max PSI)

4.25

Drift window size

400

Fairness gap

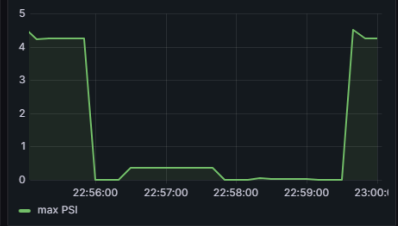
0.0376

Input drift

PSI per feature

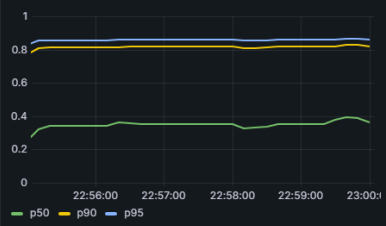


Aggregate drift score

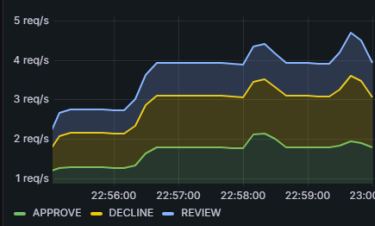


What the model is saying

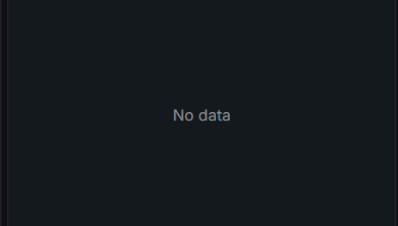
Score distribution (quantiles)



Decision mix

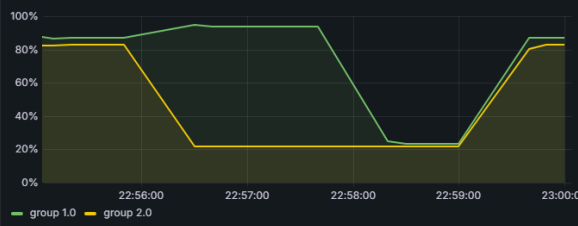


Prediction errors by type

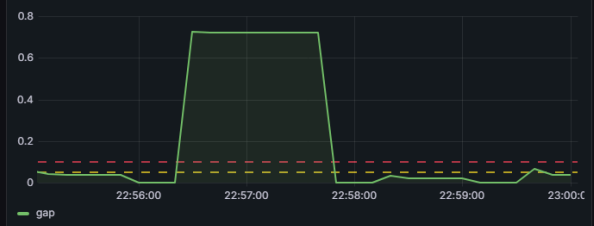


Fairness

Selection rate by group

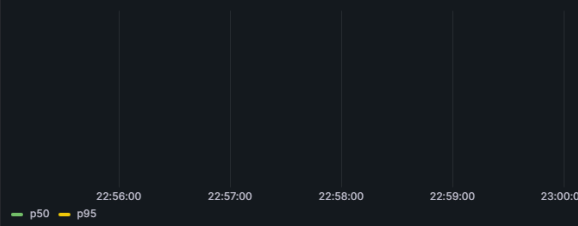


Fairness gap over time



Cost of explaining

Explanation latency



Explanations rate



Figure 2 — Model Behaviour dashboard after 400 drifted requests (strength 1.0). Drift score is red at 4.25, but fairness gap remains green at 0.038.

Under drifted traffic, the dashboard tells a dramatically different story. The drift score stat panel turns red at 4.25 — far above the 0.25 "significant" threshold. The PSI per feature panel reveals which features drove it: `payment_ratio` leads at 4.25, followed by `utilisation_ratio` at 3.09 and `LIMIT_BAL` at 2.04. This makes sense: the drift profile creates younger applicants with lower credit limits and higher utilisation, so the financial-stress features move most.

Crucially, the **fairness gap stays green at 0.038**. Both groups are equally affected by the population shift — group 1.0 at 86.9% selection rate and group 2.0 at 83.1%. The drift is broad and uniform, not targeted. The score distribution shifts upward (p50 from ~0.16 to ~0.64), and the decision mix inverts: `DECLINE` jumps from 8% to 55%.

2.3 Unfair traffic

Is the model still looking at the same world?

Drift score (max PSI)

0.353

Drift window size

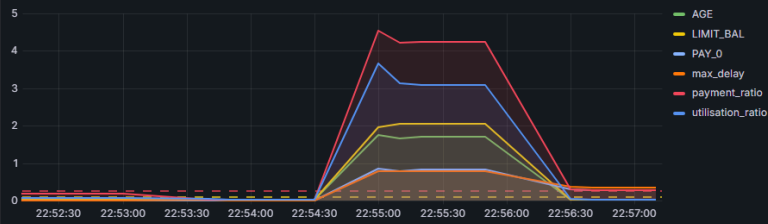
400

Fairness gap

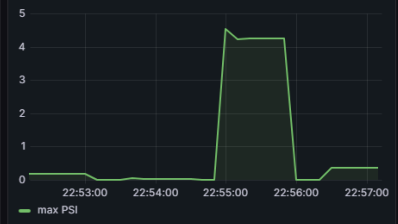
0.722

Input drift

PSI per feature

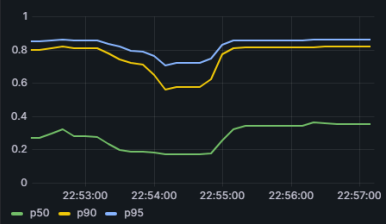


Aggregate drift score

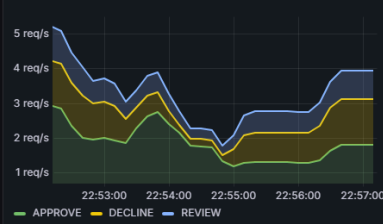


What the model is saying

Score distribution (quantiles)



Decision mix

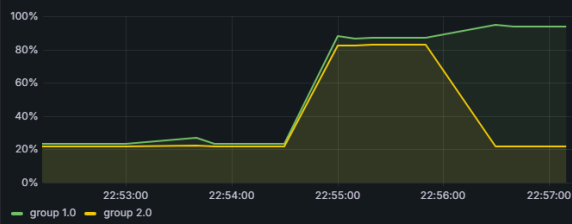


Prediction errors by type

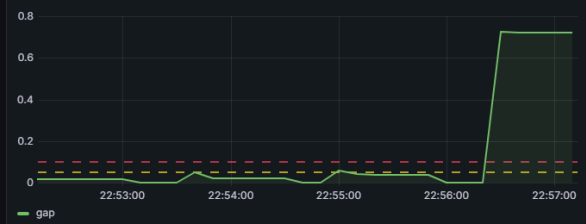
No data

Fairness

Selection rate by group

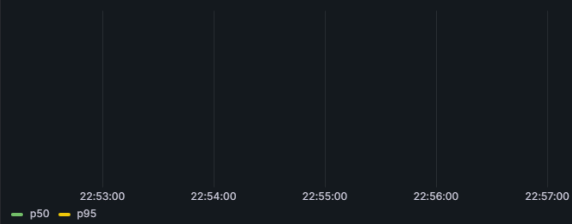


Fairness gap over time



Cost of explaining

Explanation latency



Explanations rate

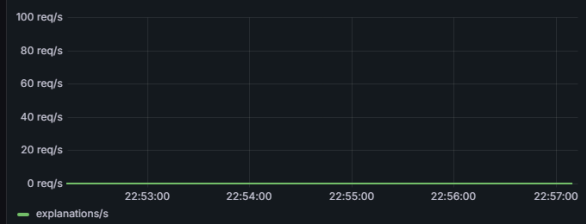


Figure 3 — Model Behaviour dashboard after 400 unfair requests. Both drift (0.353) and fairness gap (0.722) are red. The selection rate panel shows the groups diverging sharply.

Under unfair traffic, the dashboard shows a qualitatively different pattern. The drift score is 0.353 — above the 0.25 significant threshold, but only modestly so compared to the drifted run's 4.25. The **fairness gap panel is the one that explodes**, jumping to 0.722 — a 38x increase from the 0.019 baseline. The Selection Rate by Group panel makes the story visible: group 1.0's selection rate rockets to 93.8% while group 2.0 stays at 21.6%, unchanged from the normal baseline.

The per-feature PSI shows only three features moved: max_delay (0.353), payment_ratio (0.288), and PAY_0 (0.279). These are exactly the features the bias injection modified (pay_status pushed higher, pay_amt reduced to 25%). The demographic and financial features (LIMIT_BAL, AGE, utilisation_ratio) stayed at their normal-traffic PSI values — confirming the shift is targeted, not broad.

3. Analysis Questions

3.1 Which signal moved first, and by how much before anything else did?

Input drift (PSI) moved first, and it moved substantially before the decision mix or fairness gap became visible.

At mild drift (strength 0.05), the drift score was already 0.189 — inside the moderate band (0.10–0.25) — while the decision mix was essentially unchanged: 77.5% APPROVE vs 77.8% normal, and 8.5% DECLINE vs 8.0% normal. The mean score shifted by only 0.008 (from 0.2355 to 0.2437). The drift signal was **24x the noise floor** (0.189 vs the 0.030 normal baseline) at a point when no downstream metric had visibly moved.

This asymmetry is the central argument for input monitoring. PSI detects the shift in the population the model is scoring *before* that shift propagates through the model into the decisions. By the time the decline rate visibly rises, the model has been scoring the wrong population for the duration of the observation window.

The gap between when PSI moves and when the decision mix moves is the early warning window that justifies the entire drift monitoring apparatus. In this experiment, it was at least one full strength tier (0.05 → 0.15 before decline rate noticeably shifts).

3.2 Which signal would have paged you, given your own alert thresholds?

The alert rules define seven triggers. Here is which would fire under each profile:

Alert rule	Threshold	Normal	Drifted	Unfair
ModerateFeatureDrift	PSI > 0.10, 15m, warning	No	Yes	Yes
SignificantFeatureDrift	PSI > 0.25, 15m, critical	No	Yes	Yes
DriftWindowTooSmall	< 200 rows, 30m, info	No*	No*	No*
DecisionMixShift	decline > 20%, 30m, warning	No	Yes	Yes
FairnessGapWidened	gap > 0.10, 20m, critical	No	No	Yes
PredictionErrorsRising	> 0.1/s, 5m, critical	No	No	No
ExplanationLatencyHigh	p95 > 1s, 10m, warning	No	No	No

* DriftWindowTooSmall fires transiently during the initial fill (< 200 requests) but clears once the window reaches 200.

Under **drifted traffic**, three alerts would page: *SignificantFeatureDrift* (critical) fires first because PSI crosses 0.25 quickly, followed by *ModerateFeatureDrift* (warning, fires on the same data), and *DecisionMixShift* (warning, once the decline rate sustains above 20%). The **first page is SignificantFeatureDrift** — a critical-severity alert.

Under **unfair traffic**, four alerts fire. The same drift alerts trigger (PSI 0.353 > 0.25), plus *DecisionMixShift* (decline rate 32% > 20%), and critically, **FairnessGapWidened** fires — the only alert with severity: critical and signal: fairness. This is the alert that distinguishes unfair traffic from drifted traffic: it fires exclusively when the selection-rate gap between groups exceeds 0.10.

3.3 Which signal told you what was wrong, rather than only that something was? Compare the drifted and unfair runs.

The aggregate drift score cannot distinguish drifted from unfair traffic. Both runs crossed the 0.25 significant threshold (4.25 vs 0.35). The aggregate number says "*something moved*" but not "*what moved*" or "*who absorbed the change*."

Two signals separate them:

1. The per-feature PSI panel tells you WHAT moved. Under drifted traffic, all six features climb: payment_ratio (4.25), utilisation_ratio (3.09), LIMIT_BAL (2.04), AGE (1.71), PAY_0 (0.83), max_delay (0.78). This is a broad population shift — every feature participated. Under unfair traffic, only three features moved: max_delay (0.35), payment_ratio (0.29), PAY_0 (0.28). The demographic and financial features stayed at their normal baseline. A narrow shift in repayment-behaviour features, concentrated in the features the bias injection modified, points to a targeted change in one subpopulation, not a broad shift.

2. The fairness gap and selection-rate-by-group panels tell you WHO was affected.

Under drifted traffic, the fairness gap was 0.038 — both groups were equally harmed by the drift. Under unfair traffic, it jumped to 0.722: group 1.0's selection rate rose from 23.4% to 93.8% while group 2.0 stayed at 21.6%. The Selection Rate by Group panel on the dashboard makes this visually unambiguous — one line stays flat while the other skyrockets.

The aggregate drift score is an alarm. The per-feature PSI is a diagnostic. The fairness gap is an attribution. You need all three to move from "something is wrong" to "this specific subpopulation is being treated differently on these specific features." An alert system with only the aggregate would page identically for drifted and unfair traffic and give the responder no guidance on where to look — which is why the dashboard has five sections, not one number.